IP Address Classes and Subnetting Assignment

Name: _____ Class: _____

1. IP Address Classes (A, B, C, D, E)

Classful addressing is an older way to group IPv4 addresses. Modern networks usually use CIDR, but these classes are useful for learning.

Class	First-octet range	Traditional purpose	Example use
A	1–126*	Historically, very large networks	A large organization's private network may use 10.0.0.0/8 (Class A private range).
B	128–191	Historically, medium-to-large networks	A college may use private addresses from 172.16.0.0 to 172.31.255.255 (Class B p
C	192–223	Historically, smaller networks	A home router may assign 192.168.1.x addresses (Class C private range).
D	224–239	Multicast traffic	A streaming or conferencing service can use multicast groups on a managed netwo
E	240–255	Reserved/experimental; not normal host assignments	A lab may use Class E addresses in networking experiments; ordinary devices s

*0 is reserved, and 127.x.x.x is used for loopback, so the traditional Class A usable first-octet range is 1–126.

2. Find Network, Broadcast, and Valid Host Range

Given: IP address = 192.168.10.45; subnet mask = 255.255.255.0 (/24).

Step 1 — Network address: With /24, the first three octets are the network portion and the last octet is the host portion. Set the host portion to 0: **192.168.10.0**.

Step 2 — Broadcast address: Set all host bits to 1 (last octet becomes 255): **192.168.10.255**.

Step 3 — Valid host range: The network and broadcast addresses are not assigned to normal hosts. Therefore, valid host addresses are **192.168.10.1 – 192.168.10.254**.

Result: There are 254 usable host addresses in this /24 subnet.

3. Four-Subnet Plan Using 10.0.0.0/24

To create 4 equal subnets, borrow 2 host bits because $2^2 = 4$. The new prefix is /26, and the subnet mask is 255.255.255.192. Each subnet has 64 total addresses and 62 usable host addresses.

Group	Network / prefix	Subnet mask	Usable host range	Broadcast
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1–10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65–10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129–10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193–10.0.0.254	10.0.0.255

4. College Wi-Fi: Create at Least 20 Subnets from 172.16.0.0/16

Step 1: Need at least 20 subnets. Borrow 5 bits because 2■ = 16 (not enough) and 2■ = 32 (enough).

Step 2: Original prefix /16 + 5 borrowed bits = **/21**.

Step 3: /21 subnet mask = **255.255.248.0**.

Step 4: Host bits remaining = 32 – 21 = 11. Total addresses per subnet = 2^{11} = 2048. Usable hosts = 2048 – 2 = **2046** per subnet.

Answer: 32 possible subnets, with 2046 usable host addresses per subnet (under the usual IPv4 subnet rule).

5. Why 192.168.1.300 Is Invalid

An IPv4 address has four numbers (octets), and each octet must be between 0 and 255. The last number in 192.168.1.300 is 300, which is too large, so this is not a valid IPv4 address.

If the intended network is 192.168.1.0/24, valid host addresses normally look like **192.168.1.1 to 192.168.1.254**. For example, 192.168.1.30 is valid if it is not already assigned to another device. The network address is 192.168.1.0 and the broadcast address is 192.168.1.255.

Conclusion: Correct subnet masks and valid IPv4 octets are essential for devices to communicate on a network.